Real Analysis and Probability

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Appendix

Radon-Nikodým Theorem

Let $(\Omega,\ \mathcal{A})$ be a measurable space. Define

$$\mathcal{M}^+(\mathcal{A}) := \{f : \Omega \to \bar{\mathbb{R}}_+ : f \text{ is } \mathcal{A} - \text{measurable}\}.$$

For a measure μ on $(\Omega,\ \mathcal{A})$, define

$$L^2(\mu) := \{f : \Omega \to \bar{\mathbb{R}} : f \text{ is } \mathcal{A} - \text{measurable and } \int_\Omega |f(x)|^2 \mu(dx) < 0\}.$$

Theorem 0.0.1. *Let$\mu,\ \nu$ be two finite measures on $(\Omega,\ \mathcal{A})$. Then the following two statements are equivalent:*

(i). There exists a μ-almost everywhere unique $f \in \mathcal{M}^+(\mathcal{A})$ such that

$$\nu(A) = \int_A f(x)\mu(dx), \quad A \in \mathcal{A}.$$

(ii). ν is absolutely continuous with respect to μ.

Proof. $\bullet(i) \Rightarrow (ii)$ This follows from Theorem **??**.

$\bullet(ii) \Rightarrow (i)$ First, prove the uniqueness of f. Suppose$g \in \mathcal{M}^+(\mathcal{A})$also satisfies the requirement. Since ν is a finite measure, both f, g are integrable with respect to μ, and therefore finite almost everywhere. Without loss of generality, assume that f and g are real-valued. For any $A \in \mathcal{A}$,

$$\int_A (f(x) - g(x))\, \mu(dx) = 0.$$

Since $f - g$ is $\mathcal{A}$-measurable, we can take$A = \{\omega \ : \ f(\omega) - g(\omega) > 0\} \in \mathcal{A}$, which gives $\int_\Omega (f-g)^+ d\mu = 0$, and hence $f \leqslant g$ (a.s.). Similarly, $g \leqslant f$ (a.s.). Therefore,$f = g$ a.e., proving uniqueness. Now we prove that there exists an $\mathcal{A}$-measurable $f^* : \Omega \to \bar{\mathbb{R}}_+$ such that

$$\int_\Omega u d\nu \ = \int_\Omega u \cdot f^* d\mu$$

holds for all$u \in \mathcal{M}^+(\mathcal{A})$. (Taking $u = \mathbb{1}_A,\ A \in \mathcal{A}$ gives (i).)

(1). Let $\rho := \mu + \nu$, then ρ is a measure on $(\Omega,\ \mathcal{A})$. In this case, we have $L^2(\rho) = L^2(\mu) \cap L^2(\nu)$, because

$$\|.\|^2_{L^2(\rho)} = \|.\|^2_{L^2(\mu)} + \|.\|^2_{L^2(\nu)}.$$

Define

$$\Phi(u)\ := \int_\Omega u^2 d\mu + \int_\Omega (1-u)^2 d\nu, \quad u \in L^2(\rho).$$

For $u \in L^2(\rho)$, we have $\Phi(u) < \infty$, because

$$\Phi(u) \leqslant \int_\Omega u^2 d\mu + 2\int_\Omega (1+u^2) d\nu = 2\nu(\Omega) + \int_\Omega u^2 d\mu + 2\int_\Omega u^2 d\nu < \infty.$$

(2). Let

$$d^2 := \inf\{\Phi(u) : u \in L^2(\rho)\}$$

Clearly,

$$0 \leqslant d^2 \leqslant \Phi(u)\Big|_{u=0} = \int_\Omega 1^2 d\nu\ = \nu(\Omega) < \infty,$$

so d is well-defined. Note that:

$$\begin{aligned}
\Phi(u+w) + \Phi(u-w) &= \int_\Omega (u+w)^2 + (u-w)^2 d\mu + \\
&\quad + \int_\Omega (1-u-w)^2 + (1-u+w)^2 d\nu \\
&= 2\int_\Omega (u^2+w^2) d\mu + 2\int_\Omega ((1-u)^2 + w^2) d\nu \\
&= 2\Phi(u) + 2\|w\|^2_{L^2(\rho)} \qquad (0.0.1)
\end{aligned}$$

(3). Goal: d^2 can be attained. First, there exists a sequence $(f_j)_{j\in\mathbb{N}} \subset L^2(\rho)$ such that $d^2 = \lim\limits_{j\to\infty} \Phi(f_j)$. In (0.0.1), choose $u = \dfrac{1}{2}(f_j + f_k)$, $w = \dfrac{1}{2}(f_j - f_k)$. Then as$j,\ k \to \infty$:

$$\begin{aligned}
d^2 + \|\frac{f_j - f_k}{2}\|^2_{L^2(\rho)}\ &\leqslant\ \Phi(\frac{f_j + f_k}{2}) + \|\frac{f_j - f_k}{2}\|^2_{L^2(\rho)} \\
&\overset{(0.0.1)}{=} \frac{1}{2}\Phi(f_j) + \frac{1}{2}\Phi(f_k) \to d^2.
\end{aligned}$$

Thus,

$$d^2 + \limsup_{j,k\to\infty} \|\frac{f_j - f_k}{2}\|^2_{L^2(\rho)} \leqslant d^2$$

which implies that as$j,\ k \to \infty$, $\|f_j - f_k\|_{L^2(\rho)} \to 0$, meaning that $(f_j)_{j\in\mathbb{N}}$ 为 $L^2(\rho)-$ Cauchy sequence, so there exists $f \in L^2(\rho)$ such that as$j \to \infty$, $\|f_j - f\|^2_{L^2(\rho)} \to 0$, which follows from the completeness of $L^2(\rho)$.

(4). Goal: $\Phi(f) = d^2$. In (0.0.1), choose $u = f_j,\ w = f - f_j$:

$$\begin{aligned}
d^2 + \Phi(f)\ &\leqslant\ \Phi(f) + \Phi(2f_j - f) \\
&\overset{(0.0.1)}{=}\ 2\Phi(f_j) + 2\|f - f_j\|^2_{L^2(\rho)} \to 2d^2 + 0\ (j \to \infty)
\end{aligned}$$

Thus,

$$d^2 + d^2 \leqslant d^2 + \Phi(f) \leqslant 2d^2 \Rightarrow \Phi(f) \ = \ d^2.$$

(5). Goal: f is unique in $L^2(\rho)$, hence unique ρ-a.e. Suppose $f,\ g$ both achieve the minimum of $\Phi(\cdot)$. In (0.0.1), choose $u = \frac{1}{2}(f+g)$, $w = \frac{1}{2}(f-g)$:

$$\begin{aligned} d^2 + \|\frac{f-g}{2}\|^2_{L^2(\rho)} \ &\leqslant \ \Phi(\frac{f+g}{2}) + \|\frac{f-g}{2}\|^2_{L^2(\rho)} \\ &\overset{(0.0.1)}{=} \ \frac{1}{2}\Phi(f) + \frac{1}{2}\Phi(g) = \frac{1}{2}d^2 + \frac{1}{2}d^2 = d^2 \\ \Rightarrow \|f-g\|_{L^2(\rho)} \ &= \ 0 \end{aligned}$$

Thus, $f = g$ ρ-a.e. Note that:

$$\begin{aligned} (0 \vee f \wedge 1)^2 \ &\leqslant f^2, \\ (1 - 0 \vee f \wedge 1)^2 \ &\leqslant (1-f)^2. \end{aligned}$$

From

$$d^2 = \Phi(f) \geqslant \Phi(0 \vee f \wedge 1) \geqslant d^2$$

we deduce that $f = 0 \vee f \wedge 1$, since the function that minimizes $\Phi(\cdot)$ is unique. This shows that $0 \leqslant f \leqslant 1$ ρ-a.e.

(6). Let $u \in \mathcal{M}^+(\mathcal{A})$ be arbitrary and define $u_n := u \wedge n \in L^2(\rho)$. For all $t \in \mathbb{R}$:

$$\begin{aligned} 0 &\leqslant \Phi(f + t \cdot u_n) - \Phi(f) \\ &= \ \int_\Omega ((f + t\cdot u_n)^2 - f^2) d\mu + \int_\Omega ((1 - f - t\cdot u_n)^2 - (1-f)^2)\, d\nu \\ &= \ t^2 \cdot \int_\Omega u_n^2 d\mu + t^2 \cdot \int_\Omega u_n^2 d\nu \ + 2t \int_\Omega f \cdot u_n d\mu - 2t \cdot \int_\Omega (1-f)\cdot u_n d\nu \end{aligned}$$

Dividing both sides by t^2:

$$\begin{aligned} 0 &\leqslant \int_\Omega u_n^2 d\mu + \int_\Omega u_n^2 d\nu \ + \frac{2}{t}\Big(\int_\Omega f\cdot u_n d\mu - \int_\Omega (1-f)\cdot u_n d\nu \Big) \qquad (0.0.2) \\ \Rightarrow 0 \ &= \ \int_\Omega f\cdot u_n d\mu - \int_\Omega (1-f)\cdot u_n d\nu \end{aligned}$$

This is because $0 \leqslant \int_\Omega u_n^2 d\mu + \int_\Omega u_n^2 d\nu \ < \infty$ and the inequality (0.0.2) holds for all $t \neq 0$. For any $n \in \mathbb{N}$, we have

$$\int_\Omega f\cdot u_n d\mu = \int_\Omega (1-f)\cdot u_n d\nu,$$

By applying the monotone convergence theorem, we obtain

$$\int_\Omega u\cdot f d\mu \ = \ \int_\Omega (1-f)\cdot u d\nu. \qquad (0.0.3)$$

Let $u = \mathbb{1}_N$, where $N := \{f = 1\}$. Then:

$$\mu(\{f=1\}) = \int_{\{f=1\}} 1 d\mu \ = \ \int_{\{f=1\}} 0 d\nu = 0$$

$$\Rightarrow \nu(\{f = 1\}) \ = \ 0,$$

since ν is absolutely continuous with respect to μ. Let

$$\Rightarrow f^* \ := \ \frac{f}{1-f} \cdot \mathbb{1}_{N^c} \in \mathcal{M}^+(\mathcal{A})$$

For all $w \in \mathcal{M}^+(\mathcal{A})$:

$$\begin{aligned}
\int_\Omega w \cdot f^* d\mu \ &= \int_\Omega w \cdot \frac{f}{1-f} \cdot \mathbb{1}_{N^c} d\mu \\
&= \int_\Omega f \cdot \frac{w}{1-f} \mathbb{1}_{N^c} d\mu \\
&\overset{0.0.3}{=} \int_\Omega (1-f) \frac{w}{1-f} \cdot \mathbb{1}_{N^c} d\nu \\
&= \int_\Omega w \cdot \mathbb{1}_{N^c} d\nu \\
&= \int_\Omega w d\nu \ ,
\end{aligned}$$

since $\nu(N) = 0$. □

Corollary 0.0.2. *Theorem 0.0.1 also holds for a finite μ and arbitrary ν.*

Proof. Assume μ is finite and $\nu(\Omega) = \infty$. Let

$$\alpha := \sup_{\nu(B)<\infty} \mu(B) \ .$$

Then, there exists a monotonically increasing sequence $(B_n)_{n\in\mathbb{N}} \subset \mathcal{A}$ such that $\nu(B_n) < \infty$, $n \in \mathbb{N}$,and $\alpha = \lim_n \mu(B_n)$. By the continuity of measures, we get $\mu(B) = \alpha$, where $B := \cup_{n\in\mathbb{N}} B_n$. Now, define $\Omega_0 := B^c = \cap_{n\in\mathbb{N}} B_n^c$, $\Omega_1 := B_1$, $\Omega_n := B_n \backslash B_{n-1}$, $n \geqslant 2$. Then, Ω_m is the partition of Ω with the following properties:

• If A is a measurable subset of Ω_0 and $\nu(A) < \infty$, by the definition of α, we have $\mu(A) = 0$, hence $\nu(A) = 0$. So, ν takes only the values 0 or ∞ on Ω_0!

• By the definition, we have$\nu(\Omega_m) < \infty$, $m \geqslant 1$.

For $m \geqslant 1$, Theorem 0.0.1 guarantees the existence of a function f_m defined on Ω_m such that

$$\nu(E) = \int_E f_m d\mu, \quad \forall E \in \mathcal{A}, \ E \subset \Omega_m.$$

Now, define $f_0(\omega) = \infty$, $\omega \in \Omega_0$. Then, the previous equation holds for $m = 0$ as well. Now, we combine all the f_m: if $\omega \in \Omega_m$, define $f(\omega) = f_m(\omega)$. By Corollary **??**(a), the result is proven. □

Corollary 0.0.3. *Theorem 0.0.1 also holds for σ-finite μ and arbitrary ν.*

Proof. • First, there exists $(A_j)_{j\in\mathbb{N}} \subset \mathcal{A}$, $A_j \uparrow \Omega$ and $\mu(A_j) < \infty$. Note that: $A \to \mu(A \cap A_j)$ is a finite measure, so we apply Corollary0.0.2 to it.

• Define

$$h\mu(A) := \int_A h(x)\mu(dx), \quad A \in \mathcal{A}$$

where

$$h(x) := \sum_{j=1}^{\infty} \frac{2^{-j}}{1+\mu(A_j)} \cdot \mathbb{1}_{A_j}(x).$$

Then $h\mu$ and μ have the same null set, i.e., $(h > 0 \Rightarrow \frac{1}{h}$ exists $\Rightarrow \frac{1}{h}(h\mu) = \mu)$. Also, since

$$\begin{aligned}
(h\mu)(\Omega) &= \int_\Omega h(x)\mu(dx) \\
&= \sum_{j=1}^{\infty} \int_{A_j} \frac{2^{-j}}{1+\mu(A_j)} d\mu \\
&= \sum_{j=1}^{\infty} \frac{2^{-j}\mu(A_j)}{1+\mu(A_j)} \\
&\leqslant \sum_{j=1}^{\infty} 2^{-j} = 1
\end{aligned}$$

we apply Corollary 0.0.2 to ν and $h\mu$: there exists$f \in \mathcal{M}^+(\mathcal{A})$such that

$$\nu = f \cdot (h \cdot \mu) \overset{*}{=} (f \cdot h) \cdot \mu.$$

We now just need to prove $(*)$: If

$$f = \sum_{j=0}^{N} y_j \cdot 1_{B_j}$$

where $y_j \geqslant 0,\ B_j \in \mathcal{A}$, then

$$\begin{aligned}
\nu(A) &= \int_A \sum_{j=0}^{N} y_j \cdot 1_{B_j} d(h\mu) \\
&= \sum_{j=0}^{N} y_j \cdot h\mu(A \cap B_j) \\
&= \sum_{j=0}^{N} y_j \cdot \int_{A\cap B_j} h d\mu \\
&= \cdots = \int_A (f \cdot h) d\mu
\end{aligned}$$

For a general $f \in \mathcal{M}^+(\mathcal{A})$, we can apply the monotone convergence theorem to obtain the same conclusion. Hence, $(*)$holds. □

Lévy Inversion Formula

Lemma 0.0.4. $(i) \int_0^N \frac{\sin x}{x} dx \to \pi/2$ *as* $N \to \infty$.

$(ii)\ \sup_{a\geqslant 0, A\in\mathbb{R}} \Big| \int_0^a (\sin(Ax)/x)dx \Big| < \infty.$

Proof. (i) Since the function $e^{-xy}\sin x$ is integrable on $\{(x,\ y); 0 < x < a,\ 0 < y < \infty\}$, by Fubini's theorem, we have

$$\begin{aligned}
\int_0^a \frac{\sin x}{x}dx &= \int_0^a \int_0^\infty e^{-xy}\sin x dy dx \\
&= \int_0^\infty \int_0^a e^{-xy}\sin x dx dy \\
&= \int_0^\infty \big[\frac{e^{-xy}}{y^2+1}(-y\sin x - \cos x)\big]\Big|_0^a dy \\
&= \int_0^\infty \big[\{\frac{e^{-ay}}{y^2+1}(-y\sin a - \cos a)\} - \frac{-1}{y^2+1}\big]dy \\
&= \frac{\pi}{2} - \sin a \int_0^\infty \frac{ye^{-ay}}{y^2+1}dy - \cos a \int_0^\infty \frac{e^{-ay}}{y^2+1}dy. \qquad (0.0.4)
\end{aligned}$$

If $a \geqslant 1$, then

$$0 \leqslant \frac{ye^{-ay}}{y^2+1} \leqslant e^{-ay} \leqslant e^{-y} \quad \text{und} \quad 0 \leqslant \frac{e^{-ay}}{y^2+1} \leqslant e^{-y}, \qquad (0.0.5)$$

By the Dominated Convergence Theorem, we now obtain

$$\lim_{a\to\infty} \int_0^\infty \frac{ye^{-ay}}{y^2+1}dy = 0 \quad \text{und} \quad \lim_{a\to\infty} \int_0^\infty \frac{e^{-ay}}{y^2+1}dy = 0. \qquad (0.0.6)$$

Also, since $|\sin a| \leqslant 1$ and $|\cos a| \leqslant 1$, the conclusion follows from (0.0.4) and (0.0.6).

(ii) Without loss of generality, assume $a > 0$ and $A > 0$. Then

$$\begin{aligned}
\Big| \int_0^a (\sin(Ax)/x)dx \Big| &\overset{y=Ax}{=} \Big| \int_0^{aA} (\sin(y)/y)dy \Big| \\
&\overset{(0.0.4)}{\leqslant} \frac{\pi}{2} + |\sin(aA)| \int_0^\infty \frac{ye^{-aAy}}{y^2+1}dy + \int_0^\infty \frac{e^{-aAy}}{y^2+1}dy \\
&\leqslant \frac{\pi}{2} + |\sin(aA)| \int_0^\infty e^{-aAy}dy + \int_0^\infty \frac{1}{y^2+1}dy \\
&\leqslant \frac{\pi}{2} + \frac{|\sin(aA)|}{aA} + \frac{\pi}{2} \\
&\leqslant \pi + 1.
\end{aligned}$$

Thus,

$$\sup_{a\geqslant 0, A\in\mathbb{R}} \Big| \int_0^a (\sin(Ax)/x)dx \Big| \leqslant \pi + 1 < \infty.$$

□

Theorem 0.0.5. *Assume that φ is the characteristic function of the random variable X and let μ be the distribution of X. Then for any $a < b$, $a, b \in \mathbb{R}$, we have:*

$$\lim_{T\to\infty} \frac{1}{2\pi} \cdot \int_T \frac{e^{-i\theta a} - e^{-i\theta b}}{i\theta} \varphi(\theta)d\theta = \frac{1}{2}\mu\{a\} + \mu(a,\ b) + \frac{1}{2}\mu\{b\}$$

Proof. (i) If $x, y \in \mathbb{R}$ and $x \leqslant y$, then

$$\left|e^{ix} - e^{iy}\right| = \left|\int_x^y ie^{it}dt\right| \leqslant \int_x^y |ie^{it}|dt = \int_x^y 1dt = |x-y| \tag{0.0.7}$$

(ii) From Lemma 0.0.4, we obtain:

$$\lim_{T\to\infty}\int_0^T \frac{\sin(\theta\cdot(x-a))}{\theta}d\theta = \begin{cases} 0, & x = a \\ \lim_{T\to\infty}\int_0^{T\cdot(x-a)} \frac{\sin\eta}{\eta}d\eta, & x \neq a \end{cases}$$
$$= \begin{cases} 0, & x = a \\ \frac{\pi}{2}, & x > a \\ -\frac{\pi}{2}, & x < a. \end{cases}$$

(iii). Since $e^{ix} = \cos x + i\cdot\sin x$, $\forall x \in \mathbb{R}$, for any $a < b$, $a, b \in \mathbb{R}$, we have:

$$\begin{aligned} &\int_{-T}^{T} \frac{e^{i\theta(x-a)} - e^{i\theta(x-b)}}{i\theta}d\theta \\ =&\int_{-T}^{0} \frac{\mathrm{e}^{i\theta(x-a)} - e^{i\theta(x-b)}}{i\theta}d\theta + \int_0^T \frac{e^{i\theta(x-a)} - e^{i\theta\cdot(x-b)}}{i\theta}d\theta \\ =&-\int_0^{-T} \frac{e^{i\theta(x-a)} - e^{i\theta(x-b)}}{i\theta}d\theta + \int_0^T \frac{e^{i\theta(x-a)} - e^{i\theta(x-b)}}{i\theta}d\theta \\ \overset{\theta\mapsto-\theta}{=}&-\int_0^T \frac{e^{-i\theta(x-a)} - e^{-i\theta(x-b)}}{i\theta}d\theta + \int_0^T \frac{e^{i\theta(x-a)} - e^{i\theta(x-b)}}{i\theta}d\theta \\ =&2\int_0^T \frac{\sin(\theta\cdot(x-a))}{\theta}d\theta - 2\int_0^T \frac{\sin(\theta\cdot(x-b))}{\theta}d\theta \end{aligned} \tag{0.0.8}$$

From the conclusion in (ii) and the case-by-case discussion, we can deduce that when $T \to \infty$,

$$\lim_{T\to\infty}\int_T^T \frac{e^{i\theta(x-a)} - e^{i\theta(x-b)}}{i\theta}d\theta = \begin{cases} 0, & x < a \text{ oder } x > b \\ \pi, & x = a \text{ oder } x = b \\ 2\pi, & a < x < b \end{cases}$$

(iv). Now we can obtain

$$\begin{aligned} &\lim_{T\to\infty}\frac{1}{2\pi}\cdot\int_{-T}^{T} \frac{e^{-i\theta a} - e^{-i\theta b}}{i\theta}\varphi(\theta)d\theta \\ =&\lim_{T\to\infty}\frac{1}{2\pi}\cdot\int_{-T}^{T} \frac{e^{-i\theta a} - e^{-i\theta b}}{i\theta}\left(\int_{\mathbb{R}} e^{ix\theta}\mu(dx)\right)d\theta \\ \overset{(**)}{=}&\lim_{T\to\infty}\frac{1}{2\pi}\cdot\int_{\mathbb{R}}\int_{-T}^{T} \frac{e^{i\theta(x-a)} - e^{i\theta(x-b)}}{i\theta}d\theta\mu(dx) \\ \overset{(***)}{=}&\frac{1}{2\pi}\cdot\int_{\mathbb{R}}\left(\lim_{T\to\infty}\int_{-T}^{T} \frac{e^{i\theta(x-a)} - e^{i\theta(x-b)}}{i\theta}d\theta\right)\mu(dx) \\ \overset{(iii)}{=}&\int_{\mathbb{R}}\left(\frac{1}{2}\mathbb{1}_{\{a\}}(x) + \mathbb{1}_{(a,b)}(x) + \frac{1}{2}\mathbb{1}_{\{b\}}(x)\right)\mu(dx) \\ =&\frac{1}{2}\mu(\{a\}) + \mu((a,\ b)) + \frac{1}{2}\mu(\{b\}) \end{aligned}$$

For $(**)$: We can apply Fubini's theorem because

$$\int_{\mathbb{R}}\int_{-T}^{T}\Big|\frac{e^{i\theta(x-a)}-e^{i\theta(x-b)}}{i\theta}\Big|d\theta\mu(dx)\overset{(i)}{\leqslant}2T\cdot|a-b|<\infty.$$

For $(***)$: By Lemma 0.0.4(ii) and(0.0.8), there exists a constant $M>0$ such that

$$\Big|\int_{-T}^{T}\frac{e^{i\theta(x-a)}-e^{i\theta(x-b)}}{i\theta}d\theta\Big|\leqslant M<\infty.$$

Now, $(***)$ follows from the Dominated Convergence Theorem. □

Lévy Continuity Theorem

Lemma 0.0.6. *Let φ be the characteristic function of the random variable X, and μ be the distribution of X. Then for all $A>0$, we have:*

$$\mu([-2A,\ 2A])\geqslant A\Big|\int_{-1/A}^{1/A}\varphi(t)dt\Big|-1.$$

Proof. Note that

$$\begin{aligned}\frac{1}{2T}\int_{-T}^{T}\varphi(t)dt=&\frac{1}{2T}\int_{-T}^{T}\int_{\mathbb{R}}e^{itx}\mu(dx)dt\\=&\int_{\mathbb{R}}\int_{-T}^{T}\frac{1}{2T}e^{itx}dt\mu(dx)\\=&\int_{\mathbb{R}}\frac{\sin Tx}{Tx}\mu(dx)\ .\end{aligned}$$

Since $|(\sin(Tx))/Tx|\leqslant 1$ and $|\sin(Tx)|\leqslant 1$, it follows that $|(\sin(Tx))/Tx|\leqslant\frac{1}{2TA}$ for $|x|\geqslant 2A$. Thus, we have:

$$\begin{aligned}\Big|\int_{\mathbb{R}}\frac{\sin Tx}{Tx}\mu(dx)\Big|\leqslant&\mu([-2A,\ 2A])+\int_{[-2A,2A]^c}\frac{1}{2TA}\mu(dx)\\=&\mu([-2A,\ 2A])+\frac{1}{2TA}(1-\mu([-2A,\ 2A])\\=&\frac{1}{2TA}+(1-\frac{1}{2TA})\mu([-2A,\ 2A])\ .\end{aligned}$$

Choosing $T=1/A$, we obtain:

$$\Big|\frac{A}{2}\int_{-1/A}^{1/A}\varphi(t)dt\Big|\leqslant\frac{1}{2}+\frac{1}{2}\mu([-2A,\ 2A]).$$

□

Let $\mathcal{P}(\mathbb{R})$ denote the set of all probability measures on $\mathcal{B}(\mathbb{R})$, For $\mu\in\mathcal{P}(\mathbb{R})$, define

$$\hat{\mu}(t):=\int_{\mathbb{R}}e^{itx}\mu(dx)$$

as the characteristic function of μ

Theorem 0.0.7. *Let $(\mu_n)_{n\in\mathbb{N}} \subset \mathcal{P}(\mathbb{R})$. Then we have:*

(i) If $\mu_n \xrightarrow{w} \mu$, $\mu \in \mathcal{P}(\mathbb{R})$, then $\hat{\mu}_n(t) \to \hat{\mu}(t)$ converges uniformly on every finite interval.

(ii) If $\hat{\mu}_n(t) \to \varphi(t)$ for all $t \in \mathbb{R}$ and φ is continuous at 0, then:
(1) There exists $\mu \in \mathcal{P}(\mathbb{R})$ such that $\varphi = \hat{\mu}$
(2) $\mu_n \xrightarrow{w} \mu$.

Proof. (i) If $\mu_n \xrightarrow{w} \mu$, it is evident that for all $t \in \mathbb{R}$, we have $\hat{\mu}_n(t) \to \hat{\mu}(t)$. Let $\epsilon > 0$. Choose M such that $\mu([-M,\ M]^c) < \epsilon$. On $[-M,\ M]$, define a function f as 0, on $[-M-1,\ M+1]^c$ define f as 1, and linearly interpolate between them. Since $\int_{\mathbb{R}} f d\mu_n \to \int_{\mathbb{R}} f d\mu$, there exists $N \in \mathbb{N}$ such that for $n \geqslant N$, we have

$$\int_{\mathbb{R}} f d\mu_n \leqslant \epsilon,$$

Thus, we have

$$\begin{aligned}
|\hat{\mu}_n(t+h) - \hat{\mu}_n(t)| &\leqslant \int_{\mathbb{R}} |e^{ihx} - 1| \mu_n(dx) \\
&\leqslant \int_{[-M-1,M+1]} |e^{ihx} - e^{i\cdot 0}| \mu_n(dx) + \int_{[-M-1,M+1]^c} |e^{ihx} - 1| \mu_n(dx) \\
&\overset{(7)}{\leqslant} \int_{[-M-1,M+1]} |hx| \mu_n(dx) + \int_{[-M-1,M+1]^c} 2\mu_n(dx) \\
&\leqslant \int_{[-M-1,M+1]} |h|(M+1)\mu_n(dx) + 2\int_{[-M-1,M+1]^c} f\mu_n(dx) \\
&\leqslant |h|(M+1) + 2\epsilon.
\end{aligned}$$

Now for $n \geqslant N$ and $|h| \leqslant \epsilon/(M+1)$, we have

$$|\hat{\mu}_n(t+h) - \hat{\mu}_n(t)| \leqslant 3\epsilon.$$

Since $\hat{\mu}_n(\cdot)$, for $n = 1, 2, \cdots, N-1$, is uniformly continuous, there exists $\delta_0 > 0$ such that for all $|h| \leqslant \delta_0$ and $n = 1, 2, \cdots, N-1$, we have

$$|\hat{\mu}_n(t+h) - \hat{\mu}_n(t)| \leqslant 3\epsilon.$$

Let $\delta := \delta_0 \wedge \left(\frac{\epsilon}{M+1}\right)$, then for all $n \in \mathbb{N}$ and $|h| \leqslant \delta$, we have

$$|\hat{\mu}_n(t+h) - \hat{\mu}_n(t)| \leqslant 3\epsilon.$$

Let $[-K, K]$ be a finite interval with positive value K. Then there exist $t_1, t_2, \cdots, t_m \in [-K, K]$ such that $[-K, K] \subset \cup_{i=1}^m [t_i - \delta/2, t_i + \delta/2]$. Choose $N_1 \in \mathbb{N}$ such that for each $i = 1, \cdots, m$ and each $n \geqslant N_1$, we have

$$|\hat{\mu}_n(t_i) - \hat{\mu}(t_i)| \leqslant \epsilon.$$

Now let $s \in [-K, K]$ and $n \geqslant N_1$. Choose a t_i such that $|s - t_i| \leqslant \delta/2$. Then we have

$$|\hat{\mu}_n(s) - \hat{\mu}(s)| \leqslant |\hat{\mu}_n(s) - \hat{\mu}_n(t_i)| + |\hat{\mu}_n(t_i) - \hat{\mu}(t_i)| + |\hat{\mu}(t_i) - \hat{\mu}(s)| \leqslant 7\epsilon.$$

Since $\epsilon > 0$ is arbitrary, it follows that $\hat{\mu}_n(\cdot) \to \hat{\mu}(\cdot)$ converges uniformly on $[-K, K]$.

(ii) Let $\epsilon > 0$. Since $\hat{\mu}(\cdot)$ is continuous at 0 and $\hat{\mu}(0) = 1$, choose δ such that

$$\Big|\frac{1}{2\delta}\int_{-\delta}^{\delta} \hat{\mu}(t)dt - 1\Big| < \epsilon.$$

By the dominated convergence theorem, there exists $N_2 \in \mathbb{N}$ such that

$$\frac{1}{2\delta}\int_{-\delta}^{\delta} |\hat{\mu}_n(t) - \hat{\mu}(t)|dt < \epsilon$$

for $n \geqslant N_2$. Therefore, for $n \geqslant N_2$, we have

$$\Big|\frac{1}{2\delta}\int_{-\delta}^{\delta} \hat{\mu}_n(t)dt\Big| \geqslant \Big|\frac{1}{2\delta}\int_{-\delta}^{\delta} \hat{\mu}(t)dt\Big| - \frac{1}{2\delta}\int_{-\delta}^{\delta} \Big|\hat{\mu}_n(t) - \hat{\mu}(t)\Big|dt \geqslant 1 - 2\epsilon.$$

By Lemma 0.0.6, for $n \geqslant N_2$, we have

$$\mu_n[-2/\delta,\ 2/\delta] \geqslant 2(1-2\epsilon)\ -1 = 1 - 4\epsilon$$

Thus, $(\mu_n)_{n\in\mathbb{N}}$ is tight.

Let $(\mu_{n_j})_{j\in\mathbb{N}}$ be a subsequence that weakly converges, i.e., $\mu_{n_j} \xrightarrow{w} \mu$ as $j \to \infty$ for some $\mu \in \mathcal{P}(\mathbb{R})$. Then $\hat{\mu}_{n_j}(t) \to \hat{\mu}(t)$ as $j \to \infty$, so $\varphi(t) = \hat{\mu}(t)$.

Now let $(\mu_{n_l})_{l\in\mathbb{N}}$ be any subsequence. Then there exists a further subsequence $(\mu_{n_{l_k}})_{k\in\mathbb{N}}$ such that $\mu_{n_{j_k}} \xrightarrow{w} \nu \in \mathcal{P}(\mathbb{R})$ as $k \to \infty$. Therefore, $\hat{\mu}_{n_{l_k}}(t) \to \hat{\nu}(t)$ as $k \to \infty$, and hence $\hat{\nu}(t) = \varphi(t) = \hat{\mu}(t)$, so $\mu = \nu$. Therefore, $\mu_n \xrightarrow{w} \mu$ must hold. □

A Result in Complex Analysis

Lemma 0.0.8. *If $(c_n)_{n\in\mathbb{N}} \subset \mathbb{C}$ satisfies $c_n \to c \in \mathbb{C}$ as $n \to \infty$, then we have:*

$$\lim_{n\to\infty}\Big(1 + \frac{c_n}{n}\Big)^n = e^c. \tag{0.0.9}$$

Proof. Let $\ln z$ be the principal value of the logarithm for $z \in \mathbb{C}\backslash\{0\}$, i.e.,

$$\ln z = \ln|z| + i\arg z, \quad z \in \mathbb{C}\backslash\{0\}$$

For $|z| \leqslant \frac{1}{2}$, we have

$$\ln(1+z) - z = \int_0^z \frac{-w}{1+w}dw \overset{w\equiv tz}{=} -z^2\int_0^1 \frac{tdt}{1+tz}$$

Therefore, we have

$$|\ln(1+z) - z| \leqslant |z|^2 \quad \text{for} \quad |z| \leqslant \frac{1}{2}, \tag{0.0.10}$$

because $|1 + tz| \geqslant \frac{1}{2}$ holds for $|z| \leqslant \frac{1}{2}$ and $0 \leqslant t \leqslant 1$.

Since $c_n \to c \in \mathbb{C}$, for sufficiently large n, we have $|c_n/n| \leqslant \frac{1}{2}$. Thus, we obtain

$$\begin{aligned}
\Big|\ln\Big(1+\frac{c_n}{n}\Big)^n - c\Big| =& \Big|n\ln\Big(1+\frac{c_n}{n}\Big) - c\Big| \\
\leqslant& \Big|n\ln\Big(1+\frac{c_n}{n}\Big) - c_n\Big| + |c_n - c| \\
\leqslant& n\Big|\ln\Big(1+\frac{c_n}{n}\Big) - \frac{c_n}{n}\Big| + |c_n - c| \\
\overset{(0.0.10)}{\leqslant}& n\Big(\frac{c_n}{n}\Big)^2 + |c_n - c| \\
\leqslant& \frac{c_n^2}{n} + |c_n - c| \to 0, \quad \text{as } n \to \infty.
\end{aligned} \tag{0.0.11}$$

Thus, the conclusion (0.0.9) follows from the convergence in (0.0.11). □